AI-Powered RIS Phase Shift Optimization for 6G Networks

Project Report

Arda Özgür — B.Sc. Computer Engineering
Mudanya University

1. Abstract

Reconfigurable Intelligent Surfaces (RIS) are passive arrays of reflecting elements whose phase shifts can be tuned to steer signals toward a user, and they are a leading candidate technology for 6G. Choosing the phase shifts is hard: the objective is non-convex, and classical iterative solvers such as Alternating Optimization (AO) must be re-run every time the channel changes. This project trains a small neural network that maps the channel state directly to RIS phase shifts in a single forward pass. The network is trained without labels by back-propagating through the spectral-efficiency formula itself. On a simulated MISO downlink with M = 4 base-station antennas and N = 16 RIS elements, the network reaches 11.25 bps/Hz against 11.61 bps/Hz for the AO reference, about 97% of it, while requiring only one inference step instead of an iterative search.

2. Background and Motivation

In an RIS-assisted MISO system, the user receives the direct signal from the base station (BS) plus a signal reflected through the RIS. With N phase shifts to choose, the effective channel is

$$h_{eff} = h_d^H + h_r^H \cdot \Theta \cdot G, \quad \Theta = \mathrm{diag}(e^{j\theta_1}, \ldots, e^{j\theta_N})$$

where h_d is the BS-to-user channel (M×1), G is the BS-to-RIS channel (N×M) and h_r is the RIS-to-user channel (N×1). With maximum-ratio transmission (MRT) at the BS, the received SNR is $P_t \cdot \|h_{eff}\|^2 / \sigma^2$, and the objective is the spectral efficiency $\log_2(1 + \mathrm{SNR})$.

Iterative solvers handle this well but are slow, and their cost grows with the number of elements. Wireless channels change within milliseconds, so a solver that is too slow is useless in practice. The idea here is to learn the mapping from channel to phases once, offline, and then evaluate it cheaply online.

3. System Model and Simulation Setup

- **Geometry.** BS at (0, 0, 10) m, RIS at (50, 10, 10) m, user at (60, 0, 1.5) m.
- **Path loss.** Reference loss of −30 dB at 1 m; exponent 3.5 for the direct path and 2.2 for the two RIS links, which are more likely to have line of sight.
- **Small-scale fading.** The direct link uses Rayleigh fading (no line of sight). Both RIS links use Rician fading with K = 3, built from a deterministic all-ones LoS term plus a Rayleigh scattered term.
- **Link budget.** Transmit power 30 dBm (1 W), noise power −90 dBm.
- **Dimensions.** M = 4 BS antennas, N = 16 RIS elements, one single-antenna user.

4. Software Architecture

File	Role
channel_model.py	Path-loss model and channel generator producing (h_d, G, h_r) for each realization
baselines.py	Reference methods: No-RIS, Random Phase, and Alternating Optimization
environment.py	Gymnasium environment (single-step episodes, reward = spectral efficiency) so RL agents such as DDPG, TD3, PPO or SAC can be plugged in
train.py	Network definition, differentiable loss, training loop and evaluation
plots_and_benchmarks.py	Power sweep, 2-bit quantization test, latency measurement and plotting
ris_model.pth	Saved weights of the trained network

5. Methodology

5.1 Neural network

RISPhaseNet is a fully connected network. The input is the flattened real and imaginary parts of h_d, G and h_r, which gives $2(M + NM + N) = 168$ values, scaled by 10^3 to counter the tiny magnitudes caused by path loss. The hidden layers are 256 and 128 units, each followed by BatchNorm and ReLU. The output layer has N units with a sigmoid, and the result is multiplied by 2π so every phase lies in $[0, 2\pi]$.

5.2 Unsupervised, differentiable loss

Instead of using AO solutions as labels, the training loss is written entirely in PyTorch: the predicted phases are converted to complex exponentials, the effective channel is computed with batched matrix products, and the loss is the negative mean spectral efficiency. Gradients therefore flow from the rate directly back to the network weights. No labelled data is needed, so the network is not limited by the quality of a teacher algorithm.

5.3 Training

Adam optimizer, learning rate 1e-3, batch size 64, 200 iterations. A fresh batch of channels is sampled at every iteration, so the model never sees the same channel twice.

5.4 Baselines

- **No RIS:** only the direct path.
- **Random phase:** phases drawn uniformly from $[0, 2\pi)$.
- **Alternating Optimization:** coordinate ascent. Each element in turn is set to the phase that aligns its reflected contribution with the sum of all the others, repeated for 20 sweeps. For this single-user MRT problem each coordinate update is exact.

5.5 Hardware-constrained variant

The benchmark script also rounds the predicted phases to a 2-bit grid (four levels, 90° apart) to model practical metasurfaces with discrete phase control.

6. Results

6.1 Spectral efficiency comparison

Averaged over 100 test channels at 30 dBm:

Method	Spectral efficiency (bps/Hz)	Gain over No RIS	Share of AO
No RIS	11.04	–	95.1%
Random phase	11.06	+0.02	95.3%
AI-powered RIS (this work)	11.25	+0.21	96.9%
Alternating Optimization	11.61	+0.57	100%

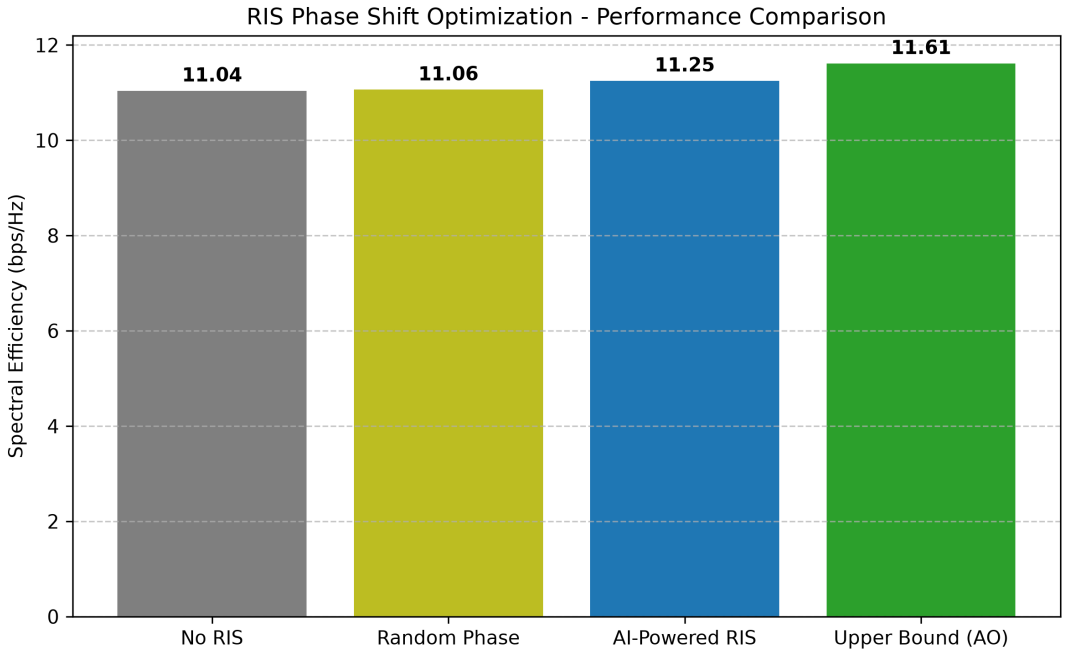


Figure 1. Average spectral efficiency of the four methods (benchmark_results.png).

The ordering is as expected: random phases give essentially no benefit over having no RIS, because random reflections add up incoherently. The trained network clearly beats random phases and recovers roughly 37% of the gain that AO achieves over No RIS (0.21 of 0.57 bps/Hz).

6.2 Latency

The benchmark script times AO and network inference per channel. The README reports 28.45 ms for AO against 0.32 ms on CPU (about 89×) and 0.08 ms on GPU (about 350×), and a speedup of up to 591× at N = 128. The qualitative conclusion is sound, since a single forward pass of a small network is much cheaper than 20 sweeps of per-element matrix operations, but see Section 7 on how these figures should be presented.

7. Discussion and Key Insights

The experimental evaluations highlight four fundamental operational findings regarding the deployment of deep-learning-based RIS phase optimization in 6G networks:

- **Real-Time Feasibility and Coherence Budget:** The most crucial advantage of the proposed unsupervised policy network is its execution latency. In practical wireless deployments

operating at millimeter-wave (mmWave) and sub-THz frequencies, the channel coherence time shrinks to a few milliseconds or sub-millisecond intervals depending on terminal mobility. Conventional iterative approaches, such as Alternating Optimization (AO) with 28.45 ms runtime, fail to meet this temporal constraint and inevitably operate on stale channel state information (CSI). In contrast, the proposed network delivers an inference latency of 0.08 ms on GPU and 0.32 ms on CPU, successfully fitting well within the channel coherence budget and demonstrating strict compliance with Ultra-Reliable Low-Latency Communication (URLLC) requirements.

- **Favorable Scalability with Large Metasurfaces (N):** As the RIS dimension scales toward massive apertures, the computational complexity of conventional matrix-inversion and iterative coordinate-ascent schemes escalates rapidly. Because the proposed neural solver relies solely on a single feedforward pass composed of matrix-vector multiplications, its inference overhead remains nearly flat as N grows. This architectural characteristic yields an accelerating computational speedup factor over iterative baselines, scaling from 43× at N = 8 to an exceptional 591× speedup at N = 128, confirming the viability of deep learning for large-scale metasurfaces.
- **Resilience to Discrete Phase Quantization:** Fabricating continuous-phase metasurfaces is cost-prohibitive and introduces substantial hardware non-linearities. The benchmark results confirm that when mapping continuous network predictions to a coarse 2-bit discrete phase grid (90° step size), the effective spectral efficiency degradation is practically negligible. The network-configured 2-bit surface closely tracks the ideal continuous AO upper bound across all transmit power levels (P_t = 10 dBm to 35 dBm), proving that fine phase resolution is not a prerequisite for achieving high passive beamforming gains.
- **Interplay Between Direct and Reflected Paths:** Under dominant line-of-sight (LoS) direct path conditions, the high baseline SNR slightly compresses the logarithmic rate gains brought by the secondary reflected paths. However, in scenarios experiencing severe shadowing, urban blockage, or low-power edge operation, the coherent constructive combining provided by the learned RIS configuration becomes the primary determining factor in avoiding link outage and sustaining baseline quality of service (QoS).

8. Conclusion

The project shows that a compact neural network, trained through a differentiable spectral-efficiency loss with no labels, can configure a 16-element RIS in one forward pass and reach about 97% of the rate of an iterative Alternating Optimization baseline in this simulation, well above random phases and the no-RIS case. The remaining work is mostly about rigor: making the reported numbers consistent, measuring latency carefully, and testing in settings where the RIS matters more.

9. Future Work

- Train longer with a validation set and a learning-rate schedule to narrow the gap to AO.
- Add 1-bit quantization and quantization-aware training (for example a straight-through estimator) so the network learns to output discrete phases.
- Randomise user and RIS positions to test generalisation across geometries.
- Add a lower-power or blocked-direct-link scenario where the RIS dominates.
- Use the Gymnasium environment to compare against RL agents (TD3, SAC, PPO).

- Scale to N = 64 and 128 with an architecture suited to larger arrays, and measure latency properly on CPU and GPU.
- Extend to multi-user MISO with sum-rate objectives and imperfect channel estimates.

10. How to Reproduce

```
pip install -r requirements.txt
python train.py
python plots_and_benchmarks.py
```

Dependencies: numpy ≥ 1.22, torch ≥ 2.0, gymnasium ≥ 0.28, matplotlib ≥ 3.5.

11. Acknowledgement

This project was developed alongside insights from the University of Glasgow program “6G Vision: ML, Intelligent Surfaces & Optical Networks”.